AI Energy Optimization (GreenAI) Documentation

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1. Problem Definition

Artificial Intelligence (AI) models consume vast amounts of energy, leading to high carbon emissions and financial costs. The AI industry lacks efficient tools to optimize software-level energy consumption without compromising performance. Most available solutions focus on hardware optimizations, leaving a gap for software-driven energy efficiency improvements.

Unmet Needs:

- **Energy-Efficient AI Development:** AI models, particularly deep learning frameworks like TensorFlow and PyTorch, are computationally intensive and require optimization to minimize energy consumption.
- **Carbon Footprint Awareness:** Developers lack visibility into how their AI models impact carbon emissions.
- **Automated Code Optimization:** There is no widely used tool that refactors AI code for improved energy efficiency without manual intervention.

Why GreenAI Exists:

GreenAI seeks to fill these gaps by developing a **machine-learning-based refactoring tool** that:

- Profiles AI code to identify high-energy-consuming methods.
- Provides detailed energy consumption metrics using Scalene and NVIDIA Nsight.
- Suggests and implements automated code optimizations to reduce energy use while preserving model performance.

Gaps in Current Solutions:

Existing profiling tools such as **CodeCarbon, Scalene, and NVIDIA Nsight** provide partial insights into AI energy consumption but lack end-to-end refactoring capabilities:

- **CodeCarbon** estimates energy usage but has inaccuracies due to profiling overhead.

- **Scalene** provides detailed CPU/GPU profiling but does not optimize energy consumption.
- **NVIDIA Nsight** focuses on GPU performance rather than energy-aware refactoring.

GreenAI aims to address these limitations by integrating **energy profiling, real-time carbon footprint tracking, and AI-driven code optimization** into a single platform.

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2. Demographics

The targeted audience of this service will be **machine learning engineers, software developers, sustainability researchers, and AI-driven companies.**

Machine Learning Engineers

Professionals involved in training and deploying AI models who will require tools to reduce energy consumption while maintaining performance.

A major advantage of GreenAI for machine learning engineers will be that it will provide **precise energy profiling and automated code refactoring**, unlike traditional profiling tools that only report raw energy usage without optimization suggestions.

Software Engineers

Developers working with AI-powered applications who will need to optimize computational efficiency.

GreenAI will enable software engineers to **maintain functionality while reducing energy consumption**, ensuring that applications run more efficiently without compromising output.

Sustainability Researchers

Researchers focused on measuring and mitigating the environmental impact of AI technologies.

GreenAI will offer **real-time carbon footprint visualization**, making it easier to quantify and track the environmental impact of machine learning workloads.

AI-Driven Companies

Organizations using AI at scale that will want to cut down on cloud computing costs and carbon emissions.

By integrating GreenAI, companies will be able to **significantly reduce operational costs and energy expenditure**, supporting their sustainability goals while maintaining computational performance.

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3. Goals

Long-Term Goals

The long-term goal of this project is to become a widely recognized tool for energy optimization in AI development by enabling machine learning engineers to write energy-efficient code. In order to achieve this, the following key objectives must be met:

- **Develop a seamless and powerful solution** that revolutionizes AI energy optimization, helping researchers and developers reduce carbon emissions while maintaining model performance.
- **Expand GreenAI's recognition** by increasing exposure through research publications, open-source collaborations, and industry partnerships.
- **Achieve adoption by major AI development teams** by making GreenAI a trusted tool in software optimization workflows.
- **Ensure financial sustainability** through potential monetization strategies such as premium features, industry consulting, or grants.

Short-Term Goals (This Semester)

In the scope of this semester, we aim to build a functional prototype of GreenAI with essential features that lay the groundwork for long-term development. Specific goals include:

Implementing Core Features:

- Utilize **Scalene/NVIDIA Nsight** to track CPU/GPU utilization of machine learning algorithms at the method level and convert that data into joules.
- Validate conversion accuracy by comparing profiling results with physical energy measurement techniques (e.g., using a power meter).

Improving Usability:

- Develop a **web-based UI** to visualize energy consumption and carbon emissions.
- Implement **real-time monitoring dashboards** that provide insights into AI model energy efficiency.

Laying the Foundation for Future Expansion:

- Integrate GreenAI with **popular AI frameworks** such as TensorFlow and PyTorch to ensure seamless adoption.
- Establish a **GitHub repository with documentation** to encourage open-source contributions and community feedback.

Relevant Tech Stack to Execute Goals

To achieve these goals, GreenAI will utilize the following technology stack:

- **Programming Languages:** Python, TypeScript
- **AI & ML Frameworks:** TensorFlow, PyTorch
- **Profiling & Optimization Tools:** Scalene, NVIDIA Nsight
- **Web Development:** React.js
- **Back-End Development:** Flask
- **Deployment & Version Control:** Docker, GitHub

How Short-Term Goals Feed into Long-Term Goals

The short-term goals lay the essential groundwork for the long-term vision of GreenAI by:

- **Creating a foundation for accurate energy profiling** – Ensuring that energy consumption data is correctly collected and analyzed is the first step toward effective optimization.
- **Developing a UI for visualization** – This will improve accessibility and usability, making it easier for researchers and developers to interpret energy usage trends.
- **Encouraging early adoption and feedback** – By integrating with widely used ML frameworks and making the tool open-source, we will engage with a community that can help refine and expand GreenAI's capabilities over time.

By accomplishing these initial objectives, GreenAI will be in a strong position to evolve into a robust energy optimization tool that is widely adopted in the AI community.

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4. Technology and Development Plans

Tech Stack Components:

Front-End Development:

- **React.js** for UI design and visualization.
- **TypeScript** for robust code quality and maintainability.

Back-End Development:

- **Python (Flask or FastAPI)** for API and server-side logic.
- **Scalene & NVIDIA Nsight** for energy profiling and method-level monitoring.
- **TensorFlow & PyTorch** for ML model optimizations.

AI & Machine Learning:

- **AI Mass Messaging (for alerts on high energy consumption).**
- **Automated Code Refactoring using AST-based transformations.**

Deployment & Version Control:

- **GitHub** for collaborative development and versioning.
- **Docker** for containerized application deployment.

Development Plan

There are several areas where the team will need to gain additional knowledge and expertise to successfully execute GreenAI. Below is a breakdown of the technologies involved, the current expertise of each team member, and the plan for learning and demonstrating mastery.

Team Members & Skill Development Plans

Ahmed:

- **Current Expertise**: Proficient in Python, familiar with TensorFlow.
- **Skills to Develop:** Scalene and NVIDIA Nsight for energy profiling.
- **Action Plan:** Study official documentation, implement small test projects, and integrate profiling tools into GreenAI.
- **Proof of Mastery:** Successfully extract and analyze energy usage data from AI models.

Layne:

- **Current Expertise:** Proficient in Java, Python, and React.js.
- **Skills to Develop:** Implementing AI-powered refactoring tools.
- **Action Plan:** Research AST-based transformations and integrate basic optimizations into test projects.
- **Proof of Mastery:** Develop a prototype that can refactor inefficient AI code.

Cristian:

- **Current Expertise:** Familiar with front-end development and API design.
- **Skills to Develop:** Building real-time dashboards for AI energy visualization.
- **Action Plan:** Learn React.js charting libraries and test different visualization approaches.
- **Proof of Mastery:** Successfully implement an interactive dashboard displaying AI energy usage.

Jake:

- **Current Expertise:** Proficient in Python, familiar with ML model optimization.
- **Skills to Develop:** Implementing AI mass messaging for notifications.
- **Action Plan:** Study natural language processing (NLP) and test different AI models for generating alerts.
- **Proof of Mastery:** Build an AI-powered messaging feature that notifies users about high energy usage.

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5. Project Plans

Our project is structured into four **sprints** to ensure incremental development and integration of energy-efficient AI solutions.

Sprint 1: Research & Project Understanding

Goals:

- Discover client expectations for this project and understand project goals
- Research possible solutions to our short-term goals and investigate existing solutions out there which tackle parts of the problem we are trying to solve.
- Record all our findings into an organized and comprehensive document for reference.

Requirement:

- Create an organized and comprehensive document outlining the goals, problem space, end-users, development plans, and project plans to communicate the team's understanding of the client's expectations.
    - Assigned to: **Jake, Ahmed, Layne and Cristian.**

Sprint 2: Profiling & Data Collection

Goals:

- Implement **Scalene/Nsight profiling** to track CPU/GPU energy usage.
- Develop a **local test environment** to validate profiling accuracy.

User Stories & Assignments:

- As a machine learning engineer, I want to profile my code to identify methods consuming high energy.
    - Assigned to: **Jake (Scalene setup), Ahmed (profiling tool integration), Layne & Cristian (test script development).**

Sprint 3: Web UI Development

Goals:

- Build a **React-based UI** to display energy consumption insights.
- Ensure **real-time visualization** of power usage trends.

User Stories & Assignments:

- As a developer, I want to see my AI model’s carbon emissions through an interactive web interface.
    - Assigned to: **Cristian & Jake (HTML/CSS skeleton), Layne & Ahmed (JS embedding and Scalene integration).**

Sprint 4: AI-Driven Optimization

Goals:

- Train a **machine learning model** to suggest energy-efficient code improvements.

User Stories & Assignments:

- As an AI developer, I want my refactoring tool to auto-generate optimization suggestions that reduce energy consumption.
    - Assigned to: Jake & Ahmed (optimization research), Layne & Cristian (ML
    - integration).
- As a software engineer developing algorithms dependent on machine learning, I want to be able to automatically refactor my code so I can reduce the energy consumption of these algorithms
    - Assigned to: Jake & Ahmed (refactorization investigation and common refactoring routines), Layne & Cristian (develop algorithms to test refactoring routines).
- As a machine learning engineer, I want the functionality of my code to remain unchanged after using a refactoring tool to reduce energy consumption so that my work is completed accurately.
    - Assigned to: Jake & Ahmed (investigate methods of preserving code functionality), Layne & Cristian (test accuracy using predefined performance metrics).

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6. Conclusion

This document outlines the key aspects of **GreenAI**, including its motivation, goals, technology stack, and development roadmap. The primary objective is to achieve the **short-term milestones** this semester while laying a strong foundation for **long-term AI energy optimization**.

By focusing on incremental sprint-based progress, we aim to deliver a functional prototype that helps **machine learning engineers** optimize their AI workloads, reducing energy costs and environmental impact. Through rigorous **profiling, visualization, and AI-driven refactoring**, GreenAI is set to revolutionize the landscape of **energy-efficient AI development**.